

Governance-First Algorithmic Systems

Final Project Contract, Submission Requirements, Evaluation Framework, and AI Governance Policy

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May 22, 2026

Abstract

This document constitutes the formal contractual framework governing the final project submission for the course on Governance-First Algorithmic Systems. The purpose of this document is to establish, with maximum clarity and precision, the methodological, operational, governance, documentation, evaluation, and reporting standards required for successful completion of the course project.

The course does not conceptualize algorithmic trading as a treasure-hunting exercise, an indicator-mining competition, or a speculative attempt to maximize isolated performance metrics through excessive optimization. The course instead approaches algorithmic trading as the disciplined construction of governed operational systems functioning under uncertainty, market stress, liquidity constraints, execution frictions, operational risks, and institutional realities.

The final project must therefore resemble a professional institutional research memorandum and governance report rather than a casual coding notebook or a collection of disconnected backtests.

Students are expected to demonstrate the ability to:

- Design an algorithmic system in a structured and disciplined manner.
- Understand and explain the QuantConnect framework architecture.
- Define and govern data usage appropriately.
- Build and justify alpha-generation logic.
- Construct portfolios consistently with strategy objectives.
- Implement and explain risk management procedures.
- Evaluate execution assumptions realistically.
- Stress-test the system under adverse historical conditions.
- Interpret performance metrics critically and honestly.
- Produce transparent and auditable documentation.
- Govern AI-assisted workflows responsibly.

The project will not be evaluated primarily on profitability. The primary objective is to evaluate whether the student understands algorithmic trading as a governed systems discipline.

1 General Philosophy of the Course

1.1 The Core Philosophy

The purpose of this course is not to train students to search for unrealistic profits through unstable optimization, excessive parameter tuning, or speculative signal mining.

The course is instead designed to help students understand how professional algorithmic systems are conceived, structured, governed, stress-tested, documented, monitored, and evaluated in institutional environments.

Professional algorithmic systems are not judged solely by profitability.

They are judged by:

- Reliability
- Governance quality
- Documentation discipline
- Stress resilience
- Transparency
- Operational realism
- Reproducibility
- Interpretability
- Risk awareness

Students are therefore expected to approach the assignment as the construction of a governed operational infrastructure rather than as a speculative trading exercise.

1.2 The Three Pillars of the Course

The course is structured around three conceptual pillars.

1.2.1 Pillar I — Algorithmic Systems as Governed Systems

Students must understand that algorithmic trading is fundamentally a systems problem.

An algorithmic strategy is not merely an indicator or a formula. It is a structured operational process composed of:

- Data
- Universe selection
- Alpha generation
- Portfolio construction
- Risk management
- Execution procedures

- Monitoring
- Stress testing
- Reporting
- Governance

The course therefore emphasizes governance, discipline, and structure from the very beginning of the strategy design process.

1.2.2 Pillar II — Research and Strategy Development

Students must understand the broad landscape of algorithmic trading methodologies.

This includes:

- Momentum systems
- Mean reversion systems
- Trend-following systems
- Carry strategies
- Relative-value systems
- Statistical arbitrage
- Cross-sectional allocation systems
- Multi-asset systems
- Regime-aware systems

The objective is not memorization but understanding the logic, assumptions, strengths, and weaknesses behind systematic trading systems.

1.2.3 Pillar III — Market-Specific Structures

Different markets possess different structural characteristics.

Students must recognize that:

- Crypto markets differ from equity markets.
- Oil futures differ from FX carry trades.
- Credit instruments differ from volatility products.
- Liquidity conditions differ across markets.
- Stress dynamics differ across asset classes.

Students are expected to demonstrate awareness of the market structure associated with the strategy they choose.

2 Contractual Nature of the Submission

2.1 Binding Nature of Requirements

This document constitutes a formal submission contract between the student and the course.

Every clause contained in this document is mandatory unless explicitly stated otherwise.

Students shall not assume that omitted information is understood or implied.

If a requirement is not explicitly documented and justified, the requirement shall be considered unmet.

2.2 Professional Standard Requirement

The final report must resemble:

- A professional institutional research report
- A governance memorandum
- A strategy review document
- An auditable operational systems report

The report must not resemble:

- An informal notebook
- A coding dump
- A marketing presentation
- A speculative narrative
- A collection of screenshots without analysis

2.3 Burden of Proof

The burden of proof lies entirely with the student.

The student must demonstrate:

- What was done
- Why it was done
- How it was implemented
- What assumptions were made
- What limitations exist
- What weaknesses remain unresolved

Assertions without evidence or justification shall not be considered valid analysis.

3 Artificial Intelligence Usage, Disclosure, and Governance

3.1 General Philosophy Regarding AI

The use of Artificial Intelligence systems is not discouraged in this course.

On the contrary, students are explicitly permitted to use AI systems as part of their research, coding, writing, debugging, analytical, documentation, and ideation workflows.

The course recognizes that modern financial research and algorithmic systems development increasingly occur in environments where AI-assisted workflows are operationally relevant.

The purpose of the course is therefore not to prohibit AI usage, but rather to evaluate whether the student can govern, supervise, validate, structure, and critically assess AI-assisted work responsibly.

The use of AI shall not negatively affect the student's grade.

The instructor expects AI usage.

The evaluation instead focuses on whether the student remained intellectually responsible for the final work product.

3.2 Human Responsibility Clause

Regardless of the level of AI participation, the student remains fully responsible for the complete contents of the submission.

The student assumes full responsibility for:

- Accuracy
- Logical consistency
- Interpretation
- Verification
- Governance quality
- Documentation quality
- Originality

The use of AI does not transfer accountability away from the student.

3.3 Mandatory AI Disclosure Requirement

Students must include a dedicated appendix titled:

Appendix — Artificial Intelligence Participation and Governance

This appendix is mandatory.

3.4 Mandatory Contents of the AI Appendix

The appendix must disclose:

- Which AI systems were used.
- Which tasks involved AI participation.
- The purpose of the AI participation.
- The extent of AI participation.
- How the student validated or supervised AI outputs.
- Which parts were primarily human-generated.
- Which parts involved collaborative human-AI workflows.

3.5 Mandatory AI Session Links

Students must provide links to the AI sessions used during the project whenever possible.

Examples include:

- ChatGPT session links
- Claude conversation links
- Notebook-based AI workflow logs

The purpose of this requirement is transparency and auditability rather than surveillance.

3.6 Creativity and Human Control

The project should demonstrate:

- Creativity
- Original synthesis
- Human judgment
- Structured reasoning
- Methodological ownership

The instructor values disciplined and creative human supervision of AI-assisted workflows.

4 Mandatory Deliverables

Students must submit the following deliverables.

Failure to provide any mandatory deliverable may render the submission incomplete.

1. Final PDF report
2. QuantConnect project link
3. Baseline backtest link
4. Stress-testing backtest links
5. Supporting charts
6. Supporting tables
7. Final compliance checklist

5 Mandatory Structure of the Report

The report must follow the exact structure specified below.

Students are strongly encouraged to use section titles identical to those listed in this document.

6 Section I — Executive Summary

6.1 Purpose of This Section

The Executive Summary should provide a concise but professional overview of the project.

6.2 What the Instructor Expects to See

The instructor expects to see:

- A clear explanation of the strategy objective.
- A concise explanation of the alpha idea.
- Identification of the market being studied.
- The time period analyzed.
- Main findings.
- Main risks and weaknesses.
- Main conclusions.

7 Section II — Universe Selection and Data Definition

7.1 Purpose of This Section

This section defines the environment in which the algorithm operates.

Weak data discipline frequently produces misleading results.

7.2 What the Instructor Expects to See

The instructor expects to see:

- A carefully defined trading universe.
- Justification for why those assets belong in the strategy.
- Evidence that the student understands liquidity and market structure.
- Clear specification of the time period and data frequency.
- Discussion of data integrity issues.

7.3 Mandatory Requirements

Students must explicitly define:

- Asset names and symbols
- Asset classes
- Exchanges or markets
- Data sources
- Time period
- Data resolution
- Price fields used
- Volume fields used

Students must discuss:

- Survivorship bias
- Look-ahead bias
- Missing observations
- Data cleaning procedures
- Data limitations

8 Section III — QuantConnect Methodology Alignment

8.1 Purpose of This Section

Students must demonstrate understanding of the QuantConnect Algorithm Framework.

The report must explicitly map the strategy into the QuantConnect architecture.

8.2 Required Framework Components

The following components must be addressed explicitly:

1. Universe Selection
2. Alpha Model
3. Portfolio Construction
4. Risk Management
5. Execution Model
6. Evaluation and Reporting

8.3 What the Instructor Expects to See

The instructor expects to see evidence that the student understands:

- How signals are generated
- How signals become positions
- How risk limits affect positions
- How orders are executed
- How portfolio construction differs from alpha generation

9 Section IV — Alpha Model and Trading Logic

9.1 Purpose of This Section

This section explains the intellectual core of the strategy.

Students must explain why the signal should theoretically work.

9.2 What the Instructor Expects to See

The instructor expects to see:

- Economic intuition
- Behavioral intuition
- Statistical reasoning
- Market structure awareness
- Logical consistency

9.3 Mandatory Comparative Analysis

Students must compare the selected strategy against at least two plausible alternatives.

Students must explain why the chosen logic was preferred.

9.4 Mandatory Variations

Students must test and discuss at least three variations of the selected logic.

The purpose of this requirement is robustness evaluation rather than optimization.

10 Section V — Portfolio Construction

10.1 Purpose of This Section

Portfolio construction explains how signals become actual portfolio allocations.

10.2 What the Instructor Expects to See

The instructor expects to see:

- Position sizing logic
- Weighting methodology
- Exposure controls
- Leverage controls
- Capital allocation reasoning

10.3 Mandatory Requirements

Students must explain:

- Weighting methodology
- Whether leverage is used
- Whether short selling is allowed
- Maximum position sizes
- Rebalancing procedures
- Cash treatment

11 Section VI — Risk Management

11.1 Purpose of This Section

Risk management is one of the most important sections of the project.

Students must demonstrate awareness that algorithmic systems can fail under stress.

11.2 What the Instructor Expects to See

The instructor expects to see:

- Drawdown controls
- Exposure controls
- Stop-loss logic
- Volatility awareness
- Portfolio-level controls

11.3 Mandatory Requirements

Students must explain:

- Maximum drawdown controls
- Volatility limits
- Position limits
- Leverage limits
- Exposure constraints

12 Section VII — Execution Model

12.1 Purpose of This Section

Execution assumptions significantly affect real-world performance.

12.2 What the Instructor Expects to See

The instructor expects to see:

- Realistic transaction cost assumptions
- Slippage awareness
- Execution timing awareness
- Liquidity awareness

12.3 Mandatory Requirements

Students must explain:

- Order types used
- Rebalance timing
- Transaction cost assumptions
- Slippage assumptions
- Liquidity assumptions

13 Section VIII — Backtesting and Evaluation

13.1 Purpose of This Section

This section evaluates the behavior of the strategy under normal conditions.

13.2 What the Instructor Expects to See

The instructor expects to see:

- Baseline backtest results
- Charts and tables
- Drawdown analysis
- Benchmark comparison
- Honest interpretation

13.3 Mandatory Performance Commentary

Students must discuss every major QuantConnect performance metric.

This includes:

- Sharpe ratio
- Sortino ratio
- Maximum drawdown
- CAGR
- Alpha
- Beta
- Volatility
- Win rate
- Turnover
- Number of trades

For each metric, students must explain:

1. What the metric means
2. Why the metric matters
3. Whether the result is acceptable
4. What weakness it reveals
5. How the strategy could improve

14 Section IX — Stress Testing

14.1 Purpose of This Section

Stress testing is one of the most important sections of the project.

Students must demonstrate that they understand how systems behave during extreme market conditions.

14.2 What the Instructor Expects to See

The instructor expects to see:

- Historically meaningful stress periods
- Properly documented stress scenarios
- Honest evaluation of weaknesses
- Evidence of resilience analysis
- Understanding of fragility

14.3 Recommended QuantConnect Stress Procedure

Students are strongly encouraged to create separate QuantConnect backtests for each stress scenario.

The recommended process is:

1. Build a baseline strategy over the full research period.
2. Select historically meaningful stress periods.
3. Create separate QuantConnect backtests restricted to those periods.
4. Maintain the same core strategy logic whenever possible.
5. Name the stress tests clearly.
6. Compare stress results against baseline results.

14.4 Examples of Stress Scenarios

Examples include:

- 2008 Global Financial Crisis
- COVID-19 market collapse
- Russia-Ukraine conflict
- Oil market collapse
- Inflation shock
- Crypto market crash

14.5 Mandatory Stress Analysis

Students must explain:

- Why the stress period matters
- What happened during the stress event
- How the strategy behaved
- Whether losses remained acceptable
- Whether the strategy remained investable
- Whether the system failed gracefully or catastrophically

15 Section X — Governance and Operational Reliability

15.1 Purpose of This Section

Students must demonstrate that they understand algorithmic systems as governed operational infrastructures.

15.2 What the Instructor Expects to See

The instructor expects to see:

- Reproducibility
- Auditability
- Transparency
- Documentation discipline
- Operational awareness

15.3 Mandatory Requirements

Students must discuss:

- Logging
- Documentation procedures
- Reproducibility
- Traceability
- Operational risks
- Model fragility

16 Section XI — Documentation and Evidence

Students must include:

- QuantConnect project link
- Baseline backtest link
- Stress-test links
- Supporting charts
- Supporting tables

All claims made in the report must be verifiable.

17 Evaluation Philosophy

The project shall not be evaluated primarily on profitability.

The primary evaluation criteria are:

- Structural rigor
- Governance quality
- Documentation quality
- Stress resilience
- Transparency
- Interpretability
- Operational realism
- Professionalism

A simple but robust strategy may receive a higher evaluation than a fragile strategy with very high returns.

18 Academic Integrity

Students must properly acknowledge:

- AI usage
- External libraries
- External research
- Third-party frameworks

Plagiarism or deceptive presentation practices constitute academic misconduct.

19 Final Compliance Checklist

General Submission

- ☐ PDF report submitted
- ☐ Report professionally formatted
- ☐ All required sections included
- ☐ Charts readable
- ☐ Tables labeled

Universe and Data

- ☐ Universe fully defined
- ☐ Asset selection justified
- ☐ Data frequency documented
- ☐ Time period documented
- ☐ Survivorship bias discussed
- ☐ Look-ahead bias discussed
- ☐ Data limitations discussed

QuantConnect Framework

- ☐ Universe Selection documented
- ☐ Alpha Model documented
- ☐ Portfolio Construction documented
- ☐ Risk Management documented
- ☐ Execution Model documented

Alpha Logic

- ☐ Trading logic explained
- ☐ Economic rationale explained
- ☐ Statistical rationale explained
- ☐ Two alternative strategies compared
- ☐ Three variations tested

Portfolio and Risk

- ☐ Position sizing documented
- ☐ Exposure controls documented
- ☐ Leverage controls documented
- ☐ Risk controls documented

Execution and Backtesting

- ☐ Slippage documented
- ☐ Transaction costs documented
- ☐ Benchmark documented
- ☐ Baseline backtest included

Stress Testing

- ☐ Stress period selected
- ☐ Stress period justified
- ☐ Stress-testing performed
- ☐ Stress results analyzed
- ☐ Failure modes discussed

AI Governance

- ☐ AI systems disclosed
- ☐ AI tasks disclosed
- ☐ AI participation explained
- ☐ Human supervision explained
- ☐ AI session links included

Evaluation

- ☐ QuantConnect statistics interpreted
- ☐ Drawdowns analyzed
- ☐ Weaknesses identified
- ☐ Improvements proposed

Documentation

- ☐ QuantConnect project link included
- ☐ Baseline backtest link included
- ☐ Stress-test links included
- ☐ Supporting evidence included

20 Final Statement

The purpose of this assignment is to help students understand algorithmic trading as a governed systems discipline rather than as speculative signal hunting.

Professional algorithmic systems are evaluated not only by returns, but by their:

- Reliability
- Transparency
- Governance quality
- Operational realism
- Stress resilience
- Documentation quality
- Auditability

Students are expected to demonstrate disciplined, structured, transparent, and intellectually honest reasoning throughout the entire project.